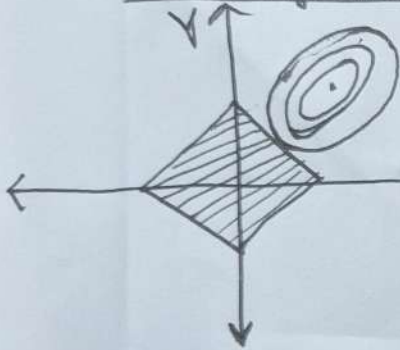


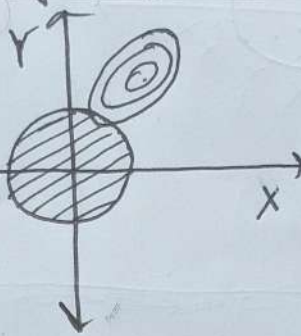
DS-413 , Assignment 3B220541) Theory Assignment

Model	Strengths	Type	Complexity	Parameters
Logistic Regression	Good for linearly separable data	Linear	Moderate	Regularisation
Linear classifier	Sensitive to outliers	Linear	Low	Non (SLS) standard least squares
Bayes classifier	Optimal if data follows assumption	Depends on distribution	Low to moderate	None ; prior assumption
SVM classifiers	High ; effective in high dimension	Linear / Non linear	High	Regularisation
Decision Tree	Good for complex decision boundaries	Non linear Hierarchical	High	Depth, splitting criterion
Linear Regression	Simple, assumes linear relationship	Linear	Low	None
Ridge / Lasso	Good for multicollinearity	Linear	Moderate	Regularisation
Decision tree Regression	Flexible but can overfit	Non linear Hierarchical	High	Depth splitting

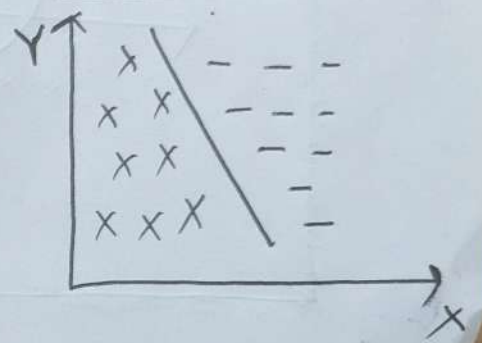
• Lasso Regression :-



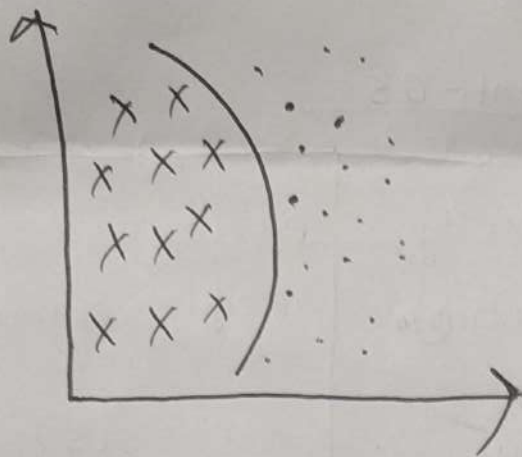
• Ridge Regression :-



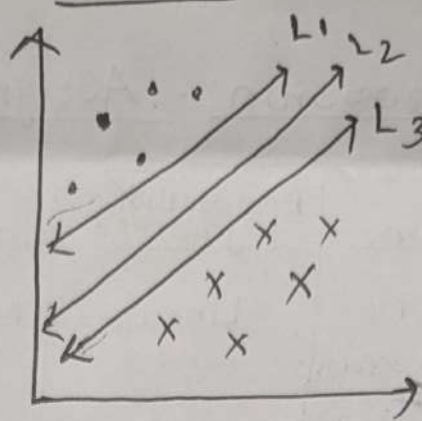
• Linear classifier :-



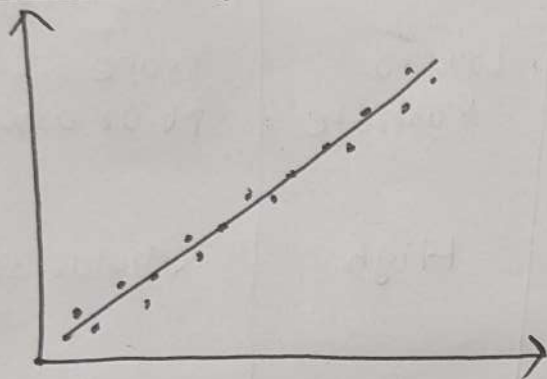
Bayes classifier:-



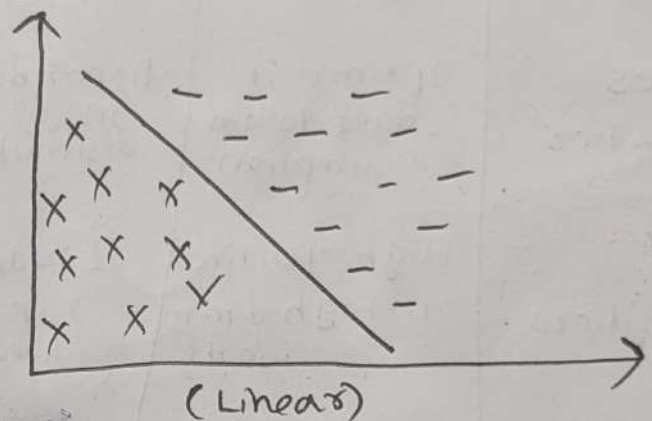
SVM :-



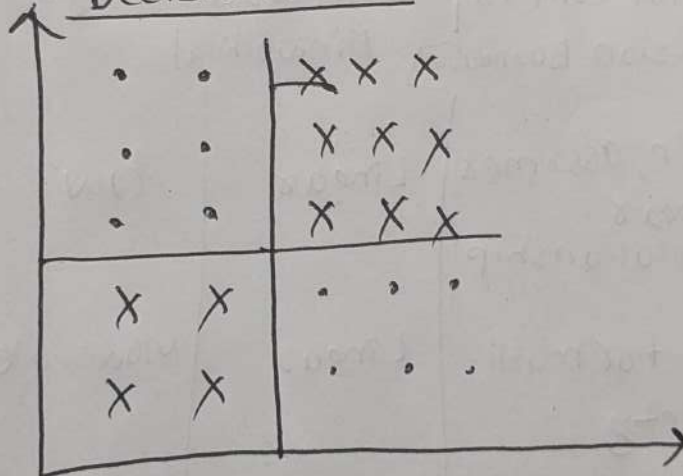
Linear Regression:-



Logistic Regression:-



Decision tree



Logistic Regression (Non-linear) :-

